Phase VII: Cleaning up

Finally, we'll need to clean up the mess we've created. Notice that the working directory has grown to a huge size, because you have three copies of each OS in it (the original ISO, the extracted files, and the files inside the *mbcd.iso*). We will dispose of the *work_dir/*, excluding *mbcd.iso* (obviously).

```
$ mv mbcd.iso ..
$ cd ..
# rm -rI work_dir
# umount /mnt/tmp?
# rmdir /mnt/tmp?
```

To add, modify or change the *mbcd* image, you do not need to mount all those distros again; just extract 'mbcd.iso', make edits, and repack it to an ISO, or use ISOmaster.

Adding more than one Knoppix-based distro

All Knoppix-based live distros have a compressed filesystem, the cloop file, in the same location — *KNOPPIX/KNOPPIX*. So no more than one Knoppix-based distro can reside in a media. To keep more than one of these, the cloop files of the second distro should be kept in different directories or have different names. After you change the cloop file's path, you need to pass it with the boot parameters shown below:

```
knoppix_dir=<path to dir containing the cloop file with leading "/">
knoppix_name=<name of the cloop file>
```

For example, we have kept the DSL cloop file in the *dsl/* directory, so we have passed *knoppix_dir=dsl/*. If the name of the cloop file is changed, then you also need to append "knoppix_name=newname" as the kernel parameter. An example for Grub is shown below. It's recommended to change only directories.

```
kernel /boot/dsl/linux24 knoppix_dir=dsl/ knoppix_name=newname
```

Notes on USB booting

To boot from a USB drive, you need to plug it in and make sure that your system supports booting from a USB drive. Consult your motherboard manual to change the boot order.

The Master Boot Record or the boot sector consists of 512 bytes and it resides at the beginning of the disk—the first sector. The first 446 bytes is reserved for the boot code. The boot code needs to be installed in order to make the BIOS boot from this drive. The next 64 bytes contain the partition table. Each partition is described with 16 bytes, so the disk can contain four primary partitions. The last two bytes contain the signature.

SYSLINUX supplies a sample MBR, the *mbr.bin* file, which can be used to overwrite the MBR boot code. To do this, first locate *mbr.bin* in your system and execute the following:

```
# dd if=/path/to/mbr.bin of=/dev/sdX
```

To clear the boot code section of MBR, execute the following command:

```
# dd if=/dev/zero of=/dev/sdX bs=440 count=1
```

Note that we didn't clear the whole 446 bytes here. This is because byte number 441 to 444 are used to store disk signatures, and the remaining two bytes are kept empty. For more information visit *en.wikipedia.org/wiki/Master_boot_record*.

It is recommended to back up your boot sector along with your files before making the USB bootable. To back up the boot sector, execute the following:

```
# dd if=/dev/sdX of=usb_mbr_bak bs=512 count=1
```

To restore the MBR, execute:

```
# dd if=usb_mbr_bak of=/dev/sdX
```

A multi-booting media is always intriguing, especially when it has a collection of great mini distros. Now that we have booted a CD/DVD and a USB drive, why don't we do the same with an internal HDD. This can be done exactly as we did for the USB drive. Keep the OS files in the root of a drive and make sure all the locations are correct in the config file; then make the HDD bootable with Grub or SYSLINUX. Next, load the config file from the menu; if Grub or SYSLINUX is already installed, then this is done—you just need to call the configuration file with the *configfile* command from the Grub shell. END

Note: Please play safe, because sometimes you only come to know you've lost something important when you restart your system the next morning.

Links and references

- Puppy Linux: *www.puppylinux.org/*
- Damn Small Linux: *www.damnsmalllinux.org/*
- GeexBox: *www.geexbox.org/*
- GoblinX: *www.goblinx.com.br/en/*
- Slitaz: *www.slitaz.org/en/*
- Austrumi: *cyti.latgola.lv/ruuni/*
- CDlinux: *www.cdlinux.info/*
- Grub: *www.gnu.org/software/grub*
- Syslinux: *www.syslinux.zytor.com*
- GRUB man pages
- SYSLINUX man pages

By: Arjun Pakrashi

The author is currently pursuing a B.Sc in computer science from Asutosh College in Kolkata. His main areas of interest are open source software, Linux programming and data structures. He plans to do research-based work, and become an OSS contributor.